

Problem Set 3

Directions

Submit two files on Moodle before the deadline:

- [1] your **solution set** as a *.pdf [2] the **R script** as a *.R type file

For both, the first line should include your student ID and the assignment number

If you worked with one or two other students write "group:" and then their names

[1] the **solution set** should include for each answer:

- the question number/part
- your answer in words
- any hand drawn diagrams
- the command in R that was used (if relevant)
- the output from Rstudio (if relevant)

[2] the **R script** should include:

- all the commands you used to generate the solutions
- individually written comments that explain your code to the grader

Driving restrictions and air quality in Mexico City

Mexico City has long suffered some of the worst air pollution in the Western Hemisphere. In response, on November 20, 1989 the city government introduced *Hoy No Circula* (HNC, "today it does not circulate"): a program that bans most drivers from using their vehicles one weekday per week based on the last digit of their license plate.

In a well-known paper, Davis (2008) examines whether HNC actually reduced air pollution. He uses hourly readings from Mexico City's network of air quality monitoring stations during the years 1986–1993, and compares pollution before and after the policy was introduced, controlling for weather, seasonality, and long-run trends. You have access to the dataset **davis2008_daily.csv** on Moodle, with daily observations from January 1, 1986 to December 31, 1993.

- **date**: the date of observation
- **year, month**: year and month of observation
- **dow**: day of the week (0 = Sunday, ..., 6 = Saturday)
- **t**: time index (number of days since January 1, 1986)
- **t2, t3, ..., t7**: powers of t (t^2 , t^3 , ... etc.)
- **hnc**: indicator (= 1 if date $\geq$ November 20, 1989, else = 0)
- For each of 5 pollutants (co, no2, o3, nox, so2):
 - **<pollutant>_mean**: daily average concentration across all monitoring stations
 - **<pollutant>_max**: daily maximum concentration across all stations
 - **log_<pollutant>**: natural log of the daily mean
- Weather (daily averages across stations):
 - **tmp**: temperature (C)
 - **rh**: relative humidity (%)
 - **wsp**: wind speed (km/hr)
 - **tmp_l1, rh_l1, wsp_l1**: lagged weather variables (1 day)

In this problem set we focus on **nitrogen oxides (NO_x)**, a group of pollutants emitted directly by motor vehicles and other combustion sources. Because cars are the main local source of NO_x in Mexico City, NO_x is arguably the most direct test of whether a driving restriction reduces air pollution.

Research question: What is the effect of the *Hoy No Circula* driving restrictions on ground level NO_x emissions in Mexico City?

1. Background & causal diagram

a. On any given weekday, HNC prevents roughly 20% of cars from being used. In your own words, why might a license-plate-based driving restriction reduce air pollution? Why might it fail to reduce air pollution?

b. Draw a causal diagram showing the relationships between: HNC, NO_x levels, weather, seasonal patterns (month-of-year and day-of-week), and long-run trends in vehicle stock. *Tip: if you are unfamiliar with the topic, you should conduct a bit of background research on what leads to particularly high vehicle emission concentrations in the ground level atmosphere of Mexico City.*

c. Why can't we simply compare average ozone before vs. after November 20, 1989 to estimate the effect of HNC?

2. Baseline model

a. Import the data into R. Estimate the baseline model and interpret $\hat{\beta}_1$.

$$\log(NO_x)_t = \beta_0 + \beta_1 HNC_t + u_t$$

b. Plot $\log(NO_x)$ over time with a vertical line at Nov 20, 1989. Describe what you see.

3. Seasonality

a. Plot the average $\log(NO_x)$ by month-of-year (across all years). Describe what you see and explain why NO_x might have this seasonal pattern.

b. Plot the average $\log(NO_x)$ by day-of-week. Describe what you see.

c. Write out and estimate the baseline model with controls for month and day-of-week. What is the effect of the policy on NO_x concentrations when we account for seasonality and weekday trends? How does your answer compare 2a? *In your answer be sure to explain in your own words what the seasonal dummies "soak up".*

4. Distributed lag in weather

a. Add contemporaneous (at time t) weather controls for temperature, humidity, wind speed to the seasonality model from question 3. Write out the new model and estimate it. Interpret the weather coefficients.

b. Now add one lag of each weather variable (tmp_l1, rh_l1, wsp_l1) and re-estimate the model from part a. Why might yesterday's weather affect today's NO_x? (*you may wish to do some background research before answering this question*) Do the coefficients on the weather variables change?

5. Adding a time trend

- a. Add a time trend to the model in 4a by including t as an explanatory variable. Write out the new model and estimate it. What is the effect of the policy on pollution? What does the time trend capture?
 - b. Construct a table showing $\hat{\beta}_1$, its standard error, and the adjusted R-squared across the four specifications from: 2a (baseline), 3c (seasonal), 4a (weather), 5a (time trend). Discuss what is happening as more controls are added.
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6. Serial correlation and HAC standard errors

- a. Why might the model in 5a suffer from serial correlation?
 - b. Using the residuals test the model in 5b for serial correlation. What do you find?
 - c. Recompute the standard error for $\hat{\beta}_1$ using Newey-West (HAC) standard errors. Does your conclusion about the effect of HNC change?
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7. Event study

The event study framework lets us estimate the effect of HNC using only pre-period data to model "normal" pollution, then comparing actual post-period pollution to the prediction.

- a. Define an estimation period and an observation period around the policy adoption date of Nov 20, 1989. Justify your choices.
- b. Estimate the model from 5a for the estimation period only, but this time drop HNC (it has no variation in the estimation period).
- c. Use the fitted model from 7b. to predict $\log(NO_x)$ over the entire sample (both estimation and observation periods). Compute the abnormal pollution for each observation:

$$AP_t = \log(NO_x)_t - \log(\widehat{NO_x})_t$$

Plot AP_t over time with a vertical line at November 20, 1989.

- d. Recall from lecture that in an event study, nonzero abnormal pollution before the event indicates *anticipation*, and nonzero abnormal pollution after indicates the *effect*. Do you see visual evidence of either?
 - e. Compare your event study estimate of the policy effect (the average of AP_t in the observation period) to $\hat{\beta}_1$ from 5a. Do the two approaches tell the same story?
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8. Discussion

In one paragraph, summarize what you have learned about the causal effect of HNC on NO_x . Given that NO_x is emitted directly by motor vehicles, what does your finding imply about the behavioral response to the driving restrictions? What is the policy implication for cities (like Delhi or Beijing) considering similar driving restrictions?